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COMPACT AND CENTRALIZED HEAT PUMP WATER HEATER SKID PACKAGE

Abstract

One embodiment provides a compact and centralized heat pump water heater system comprising a modular skid and a master controller for controlling the system. The skid includes a stratified primary storage tank for storing cold water and hot water, and an array of heat pumps for heating the cold water into the hot water. The storage tank is disposed inside the skid. The heat pumps are mounted onto an exterior of the skid. The hot water is supplied to a building structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/558,756, filed on Feb. 28, 2024, incorporated herein by reference.

TECHNICAL FIELD

[0002] One or more embodiments relate generally to water heating systems, and in particular, a compact and centralized heat pump water heater skid package for water heating applications.

BACKGROUND

[0003] Water heating is a heat transfer process that uses an energy source to heat water above its initial temperature. Example domestic and industrial uses of hot water include cooking, cleaning, bathing, space heating, etc. A water heating system is an important part of domestic dwellings such as single family homes, apartment buildings, and multi-family residential buildings, and commercial or industrial buildings such as senior living facilities, hotels, office buildings, factories, hospitals, etc. In the United States, hot water is most commonly heated with natural gas, electric resistance, or a heat pump. Electric heat pump water heaters are significantly more efficient than electric resistance water heaters.

SUMMARY

[0004] One embodiment provides a compact and centralized heat pump water heater system comprising a modular skid and a master controller for controlling the system. The skid includes a stratified primary storage tank for storing cold water and hot water, and an array of heat pumps for heating the cold water into the hot water. The storage tank is disposed inside the skid. The heat pumps are mounted onto an exterior of the skid. The hot water is supplied to a building structure.

[0005] Another embodiment provides a compact and centralized heat pump water heater system comprising a modular primary skid, a modular secondary skid, and a master controller for controlling the system. The primary skid includes a first stratified storage tank for storing cold water and hot water, and a first array of heat pumps for heating the cold water into the hot water. The first stratified storage tank is disposed inside the primary skid. The first array of heat pumps are mounted onto an exterior of the primary skid. The secondary skid includes at least one additional stratified storage tank for storing the cold water and the hot water, and a second array of heat pumps for heating the cold water into the hot water. The at least one additional stratified storage tank is disposed inside the secondary skid. The second array of heat pumps are mounted onto an exterior of the secondary skid. The hot water is supplied to a building structure.

[0006] These and other features, aspects and advantages of the present invention will become understood with reference to the following description, appended claims and accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The subject matter which is regarded as the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other objects, features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

[0008] FIG. 1A illustrates a front view of an example compact, centralized, and modular heat pump water heater (HPWH) system, in one or more embodiments;

[0009] FIG. 1B illustrates a top view of the system in FIG. 1A, in one or more embodiments;

[0010] FIG. 1C illustrates a schematic view of the system in FIG. 1A, in one or more

embodiments;

[0011] FIG. 1D illustrates a perspective view of an interior of the skid in FIG. 1A, in one or more embodiments;

[0012] FIG. 1E illustrates a perspective view of the system in FIG. 1A with more heat pumps, in one or more embodiments;

[0013] FIG. 2A illustrates a front view of an example storage tank, in one or more embodiments;

[0014] FIG. 2B illustrates a side view of the storage tank in FIG. 2A, in one or more embodiments;

[0015] FIG. 2C illustrates a top view of the storage tank in FIG. 2A, in one or more embodiments;

[0016] FIG. 3A illustrates a front view of another example compact, centralized, and modular HPWH system, in one or more embodiments;

[0017] FIG. 3B illustrates a top view of the system in FIG. 3A, in one or more embodiments;

[0018] FIG. 3C illustrates a schematic view of the system in FIG. 3A, in one or more embodiments;

[0019] FIG. 3D illustrates a top view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0020] FIG. 3E illustrates a front view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0021] FIG. 3F illustrates a right side view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0022] FIG. 3G illustrates a left side view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0023] FIG. 3H illustrates a front view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0024] FIG. 3I illustrates a right side view of the primary skid of the system in FIG. 3A, in one or more embodiments;

[0025] FIG. 3J illustrates a top view of the secondary skid of the system in FIG. 3A, in one or more embodiments;

[0026] FIG. 3K illustrates a front view of the secondary skid of the system in FIG. 3A, in one or more embodiments;

[0027] FIG. 3L illustrates a left side view of the secondary skid of the system in FIG. 3A, in one or more embodiments;

[0028] FIG. 3M illustrates a left side view of the secondary skid of the system in FIG.

[0029] 3A, in one or more embodiments;

[0030] FIG. 3N illustrates a front view of the secondary skid of the system in FIG. 3A, in one or more embodiments; and

[0031] FIG. 3O illustrates an underside of each skid of the system **200** in FIG. 3A, in one or more embodiments.

[0032] The detailed description explains the preferred embodiments of the invention together with advantages and features, by way of example with reference to the drawings.

DETAILED DESCRIPTION

[0033] One or more embodiments relate generally to water heating systems, and in particular, a compact and centralized heat pump water heater skid package for water heating applications. One embodiment provides a compact and centralized heat pump water heater system comprising a modular skid and a master controller for controlling the system. The skid includes a stratified primary storage tank for storing cold water and hot water, and an array of heat pumps for heating the cold water into the hot water. The storage tank is disposed inside the skid. The heat pumps are mounted onto an exterior of the skid. The hot water is supplied to a building structure.

[0034] Another embodiment provides a compact and centralized heat pump water heater system comprising a modular primary skid, a modular secondary skid, and a master controller for controlling the system. The primary skid includes a first stratified storage tank for storing cold

water and hot water, and a first array of heat pumps for heating the cold water into the hot water. The first stratified storage tank is disposed inside the primary skid. The first array of heat pumps are mounted onto an exterior of the primary skid. The secondary skid includes at least one additional stratified storage tank for storing the cold water and the hot water, and a second array of heat pumps for heating the cold water into the hot water. The at least one additional stratified storage tank is disposed inside the secondary skid. The second array of heat pumps are mounted onto an exterior of the secondary skid. The hot water is supplied to a building structure.

[0035] One or more embodiments of the invention may be used in different thermal energy storage systems providing heating and/or cooling in, for example, residential structures for domestic dwelling (e.g., multi-family residential buildings, etc.) or commercial/industrial structures for commercial/industrial use (e.g., office buildings, factories, etc.).

[0036] Today, challenges faced by commercial water heating systems are manifold. Some challenges include custom complex designs and installations, space constraints, environmental impact, and limited adaptability. Conventional solutions for water heating often involve intricate and time-consuming design and installation processes, escalating costs, and necessitate specialized expertise, which in turn hinders widespread implementation. Additionally, these conventional solutions typically demand substantial space or retrofitting older buildings which poses challenges in urban settings where space is at a premium. Many conventional solutions utilize refrigerants with high global warming potential (GWP), thereby contributing significantly to environmental issues. These conventional solutions often suffer from inefficiencies leading to increased energy consumption and operational costs. Moreover, the lack of modular and scalable designs in conventional solutions for water heating limit its adaptability to specific building needs, especially in multifamily residential buildings or commercial buildings.

[0037] To address these challenges, one or more embodiments of the invention provide compact, modular, and centralized heat pump water heater (HPWH) systems of varying sizes. Each HPWH system is a skid package, i.e., a self-contained unit that houses equipment and parts for heating water. Specifically, each HPWH system comprises fully modular, skid-mounted components (i.e., equipment and parts that are mounted onto a skid) that are deliverable within a single delivery as a pre-fabricated package (or system in a box) for plug-and-play. Each HPWH system is designed for factory fabrication and easy installation. Each HPWH system includes components for primary water heating and temperature maintenance such as, but not limited to, one or more heat pumps, one or more stratified storage tanks, one or more temperature maintenance tanks, one or more all pumps, one or more expansion tanks, one or more mixing valves, one or more power landings, and one or more controls. All components of a HPWH system are mounted onto or housed within one or more skids, thereby reducing footprint and upfront costs including shipping expenses; this efficient arrangement utilizes one or more heat pumps with high heating capacity and minimal greenhouse gas emissions. As described in detail later herein, each HPWH system is designed to ensure optimal operation with multiple heat pumps connected in parallel, enhancing redundancy and reliability. For example, in one embodiment, a HPWH system is capable of generating heat ranging from 31 kBtu to 185 kBtu.

[0038] Interested customers can select a desired HPWH system from a catalog of pre-engineered and fully-assembled HPWH systems of different sizes. The HPWH systems available for selection are versatile, catering to a variety of building sizes, and addressing key market concerns of installation complexity, space efficiency/constraints, and environmental impact. The HPWH systems available for selection allow for greater flexibility and adaptability, making it suitable for a diverse range of building sizes and requirements. A distributor that a desired HPWH system is ordered from can deliver on-site the desired HPWH system. Each HPWH system's comprehensive, all-in-one design simplifies installation, requiring only plug-and-play with water and power connections available on-site. Furthermore, unlike many conventional solutions for water heating, each HPWH system utilizes low GWP refrigerants, thereby minimizing the system's environmental

footprint.

[0039] The utility of each HPWH system extends beyond its compactness and eco-friendliness. Each HPWH system's pre-engineered design eliminates the need for customized engineering and system design which in turn significantly reduces time and costs associated with conventional custom installations. This streamlined approach allows for ease of installation and reliable operation, making it accessible to a wider range of users, including those without specialized knowledge of water heating systems. Due to its ability to overcome common challenges faced by conventional solutions for water heating, each HPWH system simplifies the design process, simplifies the installation process, reduces physical space requirements, lowers greenhouse gas emissions with use of eco-friendly refrigerants, and provides a versatile solution adaptable to various building sizes. Each HPWH system's pre-engineered, easy-to-install nature ensures consistent, reliable operation, addressing the complexities and inefficiencies often associated with conventional water heating systems.

[0040] One or more embodiments of the invention provide a drop-in modular HPWH system that is scalable to any building size and that is fossil fuel free.

[0041] FIG. 1A illustrates a front view of an example compact, centralized, and modular heat pump water heater (HPWH) system **100**, in one or more embodiments. FIG. 1B illustrates a top view of the system **100** in FIG. 1A, in one or more embodiments. FIG. 1C illustrates a schematic view of the system **100** in FIG. 1A, in one or more embodiments. As shown in FIGS. 1A-1C, all components (i.e., equipment and parts) of the system **100** are mounted onto or housed within a single skid **110**. In one embodiment, the skid **110** is an enclosure comprising a steel frame **50** (FIG. 1D) with removable insulated side panels (e.g., side panels **110A**, **110B**, and **110T**, etc.).

[0042] In one embodiment, the components of the system **100** include, but are not limited to, the following: (1) an array of heat pumps **120** for heating water, (2) one power connection panel **130** for power landing, (3) one stratified primary storage tank **140** for storing water, (4) one swing tank **150** for maintaining a setpoint temperature (e.g., 120° F.) in a hot water recirculation loop **190**, (5) one expansion tank **160** for preventing excessive water pressure as the system **100** heats up water, (6) a plurality of water connections **170** for exchanging water with a building the system **100** is located at, and (7) a plurality of piping connections **180** for drainage.

[0043] In one embodiment, each heat pump **120** is a carbon dioxide (CO₂) heat pump that uses CO₂ as a refrigerant. In one embodiment, each heat pump **120** allows for hot water production even when ambient temperatures fall to -20° F. or below.

[0044] In one embodiment, the array of heat pumps **120** is mounted onto an exterior of the skid **110**. For example, in one embodiment, the heat pumps **120** are arranged and distributed across an exterior of a front side panel **110A** of the skid **110**, as shown in FIG. 1A.

[0045] In one embodiment, the array of heat pumps **120** include, but is not limited to, four (4) heat pumps **120**. In another embodiment, the array of heat pumps **120** include a different number of heat pumps **120** instead (e.g., six (6) heat pumps **120**, as shown in FIG. 1E).

[0046] In one embodiment, the power connection panel **130** is mounted onto an exterior of the skid **110**. For example, in one embodiment, the power connection panel **130** is mounted onto an exterior of a left side panel **110B** of the skid **110**, as shown in FIG. 1A.

[0047] FIG. 1D illustrates a perspective view of an interior of the skid **110** in FIG. 1A, in one or more embodiments. For illustration purposes, some side panels of the skid **110** are removed to show the interior of the skid **110** in FIG. 1D. The system **100** is fitted into the frame **50** and mounted onto the skid **110**.

[0048] As shown in FIG. 1C-1D, in one embodiment, the storage tank **140**, the swing tank **150**, and the expansion tank **160** are housed inside of the skid **110**. As further shown in FIG. 1C, the plurality of water connections **170** (FIGS. 1A-1B) include a hot water recirculation return **170A** (e.g., a 1½ inch copper stub) for returning hot water to the system **100**, a cold water inlet **170B** (e.g., a 2 inch copper stub) for receiving cold water (e.g., 55° F.) from the building, and a hot water outlet **170C**

(e.g., a 2 inch copper stub) for supplying hot water to the building.

[0049] In one embodiment, the storage tank **140** includes a cold water inlet port **143** for receiving cold water (e.g., from the cold water inlet **170B**). The storage tank **140** includes a heat pump supply port **146** for supplying cold water stored (towards the bottom of the storage tank **140**) to the heat pumps **120** for heating. Each heat pump **120** includes a cold water inlet **120A** for drawing cold water (e.g., from the storage tank **140**) which it then heats. Each heat pump **120** includes a hot water outlet **120B** for supplying hot water to the storage tank **140** for storage.

[0050] In one embodiment, the storage tank **140** includes a heat pump return port **144** for receiving hot water from the heat pumps **120** which it then stores. The storage tank **140** includes a hot water outlet port **145** for supplying hot water stored (towards the top of the storage tank **140**) to the swing tank **150**.

[0051] In one embodiment, the swing tank **150** includes a hot water inlet port **151** for receiving hot water (e.g., from the storage tank **140** or the hot water recirculation loop **190**) which it maintains the setpoint temperature of (e.g., 120° F.). The swing tank **150** includes a hot water outlet port **152** for supplying hot water (e.g., either to the building via the hot water outlet **170C** or for recirculation via the hot water recirculation loop **190**).

[0052] In one embodiment, the system **100** includes one or more temperature sensors **141** (i.e., temperature probes). For example, in one embodiment, the storage tank **140** includes at least one temperature sensor **141** disposed within a thermowell **142** (FIG. 2A) of the storage tank **140**. In one embodiment, the system **100** includes one or more flow monitors to monitor pressure as water flows through the system **100**.

[0053] In one embodiment, each tank **140**, **150** includes a temperature and pressure relief port **147**, **153** for relieving an excessive amount of pressure or heat in the tank **140**, **150**, thereby protecting the tank **140**, **150** from any damage the excessive amount might have caused.

[0054] In one embodiment, the piping connections **180** include, but are not limited to, the following: a heat pump condensate drain (e.g., heat pump condensate drain **290B** in FIG. 3H, heat pump condensate drain **295B** in FIG. 3N) (e.g., a ½ inch PVC stub) for drainage of condensation from the heat pumps **120**, and tank temperature and pressure drains (e.g., tank temperature and pressure drains **290A** and **290C** in FIG. 3H, tank temperature and pressure drains **295A** and **295C** in FIG. 3N) (e.g., ¾ inch copper stubs) for drainage of any water leaks from each tank **140**, **150** after all excess air escapes from the tank **140**, **150** via its temperature and pressure relief port **147**, **153**.

[0055] In one embodiment, the system **100** includes a master mixing valve **191**, one or more balancing valves **193**, one or more check valves **194**, and a recirculation pump **192**. Together with the swing tank **150**, the master mixing valve **191** and the recirculation pump **192** help to maintain the setpoint temperature (i.e., target temperature) of the water in the hot water recirculation loop **190** to recover standby losses during periods of low hot water usage in the building. The swing tank **150** keeps primary storage of water in the storage tank **140** separate from the hot water recirculation loop **190**, and provides thermal storage to maintain the setpoint temperature. In one embodiment, the swing tank **150** includes an electrical element that maintains the setpoint temperature during extended periods of no water usage in the building. The swing tank **150** increases the efficiency of the heat pumps **120** by ensuring that the heat pumps **120** only heat cold water. Further, the swing tank **150** provides extra capacity beyond the capabilities of the heat pumps **120** for extreme draws (i.e., extreme water usage in the building).

[0056] Containing the system **100** within the skid **110** allow for easy delivery and installation of a large-scale water heating system, i.e., the skid **110** allows the system **100** to be easily transported and integrated. The skid **110** is delivered pre-engineered and pre-packaged to the site of the building when the skid **110** is ready to be installed. In one embodiment, an exterior of a top side panel **110T** of the skid **110** includes one or more lift points **111** (e.g., rigging eyelets with substantially about 1 inch through holes). For example, in one embodiment, each corner of the

exterior of the top side panel **110T** includes a lift point **111** positioned within proximity of the corner, as shown in FIG. **1B**. The lift points **111** allow the entire skid **110** to be rigged such that a crane or a forklift can lift the skid **110** and drop-in the skid **110** into position. The skid **110** may be positioned onto a flat surface, such as on the rooftop of the building or on the ground adjacent to the building. In one embodiment, one or more tie down points **112** extend from an underside **110U** of the skid **110** for securing the skid **110** onto the surface. As shown in FIG. **1B**, in one embodiment, the tie down points **112** include corner base plates **112A** at corners of the underside **110U**, side mounts **112B** at a first pair of opposing sides of the underside **110U**, and side mounts **112C** at a second pair of opposing sides of the underside **110U**. Each corner base plate **112A** and each side mount **112B**, **112C** is secured to the surface via anchor bolts.

[0057] In one embodiment, the skid **110** is a fully insulated enclosure (i.e., the steel frame **50** (FIG. **1D**) includes the insulated side panels **110A**, **110B**, **110T**, etc.), thereby allowing the system **100** to perform better than conventional water heating solutions, such as water heaters placed in mechanical rooms.

[0058] In one embodiment, the system **100** includes a master controller with wireless communication options (e.g., WiFi, LTE, BMS). For example, in one embodiment, the master controller is integrated in or coupled to the power connection panel **130**. As another example, in one embodiment, the master controller is integrated in or coupled to a heat pump control terminal box (e.g., heat pump control terminal box **260** or **270** in FIG. **3A**) for controlling the array of heat pumps **120**.

[0059] After the skid **110** is secured, the water connections **170** are connected to the building's water lines, the power connection panel **130** is connected to the building's main power supply, and the master controller is powered on to begin running and starting the system **100**. The system **100** is a skid package that provides a plug-and-play drop-in hot water solution.

[0060] FIG. **1E** illustrates a perspective view of the system **100** in FIG. **1A** with more heat pumps **120**, in one or more embodiments. In another embodiment, the array of heat pumps **120** includes six (6) heat pumps **120**, instead of four (4) heat pumps **120**. The heat pumps **120** are arranged and distributed across the exterior of the front side panel **110A** of the skid **110**, as shown in FIG. **1E**.

[0061] Any dimensions shown in FIGS. **1A-1B** are provided as examples only; the dimensions, size, and/or thickness of the skid **110** and/or other components of the system **100** are not limited to the dimensions shown in these figures. In one embodiment, a height of the skid **110** is substantially about 103¼ inches, as shown in FIG. **1A**. In one embodiment, a length of the skid **110** ranges from substantially about 144 inches to substantially about 149 inches, as shown in FIG. **1A**. In one embodiment, a width of the skid **110** ranges from substantially about 75¾ inches to substantially about 94 inches, as shown in FIG. **1B**.

[0062] In one embodiment, the skid **110** includes a service access door (e.g., service access door **212** or **222** in FIG. **3B**) providing a user (e.g., a service technician) with access to components inside the skid **110**.

[0063] FIG. **2A** illustrates a front view of an example storage tank **140**, in one or more embodiments. FIG. **2B** illustrates a side view of the storage tank **140** in FIG. **2A**, in one or more embodiments. FIG. **2C** illustrates a top view of the storage tank **140** in FIG. **2A**, in one or more embodiments. In one embodiment, a storage tank **140** of a HPWH system **100** includes a plurality of thermowells **142** distributed along the tank **140**, wherein each thermowell **142** is shaped to receive and maintain a temperature sensor **141** (FIG. **1C**). For example, in one embodiment, the storage tank **140** includes four thermowells **142** spaced out vertically (e.g., about 14 inches) across the tank **140**.

[0064] In one embodiment, the storage tank **140** includes the cold water inlet port **143** for receiving cold water, the heat pump return port **144** for receiving hot water from the heat pumps **120**, the hot water outlet port **145** for supplying hot water from the tank **140**, the heat pump supply port **146** for supplying cold water from the tank **140** to the heat pumps **120**, and the temperature and pressure

relief port **147** for relieving an excessive amount of pressure or heat in the tank **140**. The storage tank **140** further includes a tank water inlet port **148** for receiving water from another storage tank **140** that is interconnected to the tank **140** (e.g., multiple storage tanks **140** interconnected/ganged together, as shown in FIG. 3C).

[0065] The storage tank **140** provides on demand supply for heat pumps **120**. In one embodiment, the storage tank **140** stores water at 150° F. for higher effective volume.

[0066] A size of the storage tank **140** varies depending on water demand/usage of a building. For example, a size of the storage tank may be, but is not limited to, 175 gallons, 220 gallons, 275 gallons, 325 gallons, 450 gallons, or 500 gallons.

[0067] Any dimensions shown in FIGS. 2A-2B are provided as examples only; the dimensions, size, and/or thickness of the storage tank **140** are not limited to the dimensions shown in these figures. In one embodiment, spacing between the thermowells **142** is substantially about 14 inches, as shown in FIG. 2A. In one embodiment, spacing between the last/bottommost thermowell **142** and the heat pump supply port **146** is substantially about 21 inches, as shown in FIG. 2A. In one embodiment, spacing between the cold water inlet port **143** and the heat pump supply port **146** is substantially about 12 $\frac{7}{8}$ inches, as shown in FIG. 2A. In one embodiment, a height of the storage tank **140** from its top to the heat pump supply port **146** is substantially about 75 inches, as shown in FIG. 2B. In one embodiment, spacing between the heat pump return port **144** and the heat pump supply port **146** is substantially about 64 inches, as shown in FIG. 2B.

[0068] In one embodiment, multiple skids (e.g., skids **210** and **220** in FIG. 3A) are combined to create a large-scale HPWH system (e.g., system **200** in FIG. 3A) scalable to any building size and/or building water demand/usage.

[0069] FIG. 3A illustrates a front view of another example compact, centralized, and modular heat pump water heater (HPWH) system **200**, in one or more embodiments. FIG. 3B illustrates a top view of the system **200** in FIG. 3A, in one or more embodiments. FIG. 3C illustrates a schematic view of the system **200** in FIG. 3A, in one or more embodiments. As shown in FIGS. 3A-3C, all components of the system **200** are mounted onto or housed within two skids—a primary skid **210** and a secondary skid **220**. In one embodiment, each skid **210**, **220** is an enclosure comprising a steel frame (e.g., frame **50** in FIG. 1D) with removable insulated side panels (e.g., side panels **210A**, **210B**, **210C**, **210T**, **220A**, **220B**, **220C**, **220T**, etc.). The system **200** is fitted into the frames of the skids **210** and **220** and mounted onto the skids **210** and **220**.

[0070] In one embodiment, the components of the system **200** that are mounted onto or housed within the primary skid **210** include, but are not limited to, the following: (1) a first array of heat pumps **120** for heating water, (2) a first power connection panel **230** for power landing, (3) a first stratified primary storage tank **140** (e.g., STORAGE TANK 1 in FIG. 3C) for storing water, (4) one swing tank **150** for maintaining a setpoint temperature (e.g., 120° F.) in a hot water recirculation loop **296** (FIG. 3C), (5) one expansion tank **160** for preventing excessive water pressure as the system **200** heats up water, (6) a plurality of water connections **250** for exchanging water with a building the system **200** is located at, (7) a plurality of piping connections **290** for drainage, (8) a first heat pump control terminal box **260** for controlling the first array of heat pumps **120**, and (9) a plurality of inter-skid piping connections **280** for connecting with one or more components mounted onto or housed within the secondary skid **220**.

[0071] In one embodiment, the components of the system **200** that are mounted onto or housed within the secondary skid **220** include, but are not limited to, the following: (1) a second array of heat pumps **120** for heating water, (2) a second power connection panel **240** for power landing, (3) at least a second stratified primary storage tank **140** and a third stratified primary storage tank **140** (e.g., STORAGE TANK 2 and STORAGE TANK 3 in FIG. 3C) for storing water, (4) a plurality of piping connections **295** for drainage, (5) a second heat pump control terminal box **270** for controlling the second array of heat pumps **120**, and (6) a plurality of inter-skid piping connections **285** for connecting with one or more components mounted onto or housed within the primary skid

210.

[0072] As described in detail later herein, the system **200** is scalable using multiple storage tanks **140** across the skids **210** and **220** that are interconnected/ganged together.

[0073] In one embodiment, each heat pump **120** is a CO.sub.2 heat pump that uses CO.sub.2 as a refrigerant. In one embodiment, each heat pump **120** allows for hot water production even when ambient temperatures fall to -20° F. or below. In one embodiment, the first array of heat pumps **120** includes, but is not limited to, four (4) heat pumps **120** mounted onto an exterior of the primary skid **210** (e.g., arranged and distributed across an exterior of a front side panel **210A** of the primary skid **210**), and the second array of heat pumps **120** includes, but is not limited to, four (4) heat pumps **120** mounted onto an exterior of the secondary skid **220** (e.g., arranged and distributed across an exterior of a front side panel **220A** of the secondary skid **220**). In another embodiment, each array of heat pumps **120** includes a different number of heat pumps (e.g., the second array includes six (6) heat pumps **120**, whereas the first array includes four (4) heat pumps **120**, as shown in FIG. 3C).

[0074] In one embodiment, the second power connection panel **240** is mounted onto an exterior of the secondary skid **220** (e.g., on an exterior of a side panel of the secondary skid **220**, such as a right side panel **220C**, as shown in FIG. 3N).

[0075] As shown in FIG. 3C, in one embodiment, the first storage tank **140**, the swing tank **150**, and the expansion tank **160** are housed inside the primary skid **210**. As further shown in FIG. 3C, the plurality of water connections **250** (FIGS. 3A-3B) include a hot water recirculation return **250A** (e.g., a 1½ inch copper stub) for returning hot water to the system **200**, a cold water inlet **250B** (e.g., a 2 inch copper stub) for receiving cold water from the building, and a hot water outlet **250C** (e.g., a 2 inch copper stub) for supplying hot water to the building.

[0076] In one embodiment, the second storage tank **140** (STORAGE TANK 2) in the secondary skid **220** includes a cold water inlet port **143** for receiving cold water from the cold water inlet **250B**, via the inter-skid piping connections **280**, **285**. The second storage tank **140** includes a heat pump supply port **146** for supplying cold water stored (towards the bottom of the second storage tank **140**) to: (1) the first array of heat pumps **120** on the primary skid **210**, via the inter-skid piping connections **280**, **285**, for heating, and (2) the second array of heat pumps **120** on the secondary skid **220** for heating. Each heat pump **120** includes a cold water inlet **120A** for drawing cold water from the second storage tank **140** which it then heats. Each heat pump **120** includes a hot water outlet **120B** for supplying hot water to the first storage tank **140** (STORAGE TANK 1) in the primary skid **210** for storage (via the inter-skid piping connections **280**, **285** for heat pumps on the secondary skid **220**).

[0077] In one embodiment, the first storage tank **140** (STORAGE TANK 1) in the primary skid **210** includes a heat pump return port **144** for receiving hot water from the heat pumps **120** which it then stores. The first storage tank **140** includes a hot water outlet port **145** for supplying hot water stored (towards the top of the first storage tank **140**) to the swing tank **150** in the primary skid **210**.

[0078] In one embodiment, the second storage tank **140** (STORAGE TANK 2) in the secondary skid **220** includes a hot water outlet port **145** for supplying hot water stored in the second storage tank **140** to the third storage tank **140** (STORAGE TANK 3) in the secondary skid **220**. The third storage tank **140** includes a tank water inlet port **148** for receiving water from the second storage tank **140**. The third storage tank **140** includes a hot water outlet port **145** for supplying hot water stored in the third storage tank **140** to the first storage tank **140** (STORAGE TANK 1) in the primary skid **210**, via the inter-skid piping connections **280**, **285**.

[0079] In one embodiment, the first storage tank **140** (STORAGE TANK 1) in the primary skid **210** includes a tank water inlet port **148** for receiving water from the third storage tank **140** in the secondary skid **220**, via the inter-skid piping connections **280**, **285**.

[0080] In one embodiment, the swing tank **150** in the primary skid **210** includes a hot water inlet port **151** for receiving hot water (e.g., from the first storage tank **140** or the hot water recirculation

loop **296**) which it maintains the setpoint temperature of. The swing tank **150** includes a hot water outlet port **152** for supplying hot water (e.g., either to the building via the hot water outlet **250C** or for recirculation via the hot water recirculation loop **296**).

[0081] In one embodiment, the system **200** includes one or more temperature sensors **141** (i.e., temperature probes). For example, in one embodiment, each storage tank **140** includes at least two temperature sensors **141** disposed within thermowells **142** (FIG. 2A) of the storage tank **140**, the swing tank **150** includes at least one temperature sensor **141**, and multiple temperature sensors **141** are positioned along the hot water recirculation loop **296**. In one embodiment, the system **200** includes one or more flow monitors to monitor pressure as water flows through the system **200**.

[0082] In one embodiment, each tank **140**, **150** includes a temperature and pressure relief port **147**, **153** for relieving an excessive amount of pressure or heat in the tank **140**, **150**, thereby protecting the tank **140**, **150** from any damage the excessive amount might have caused.

[0083] In one embodiment, the system **200** includes a master mixing valve **297**, one or more balancing valves **298**, one or more check valves **299**, and a recirculation pump **294**. Together with the swing tank **150**, the master mixing valve **297** and the recirculation pump **294** help to maintain the setpoint temperature (i.e., target temperature) of the water in the hot water recirculation loop **296** to recover standby losses during periods of low hot water usage in the building. The swing tank **150** keeps primary storage of water in the storage tanks **140** separate from the hot water recirculation loop **296**, and provides thermal storage to maintain the setpoint temperature. In one embodiment, the swing tank **150** includes an electrical element that maintains the setpoint temperature during extended periods of no water usage in the building. The swing tank **150** increases the efficiency of the heat pumps **120** by ensuring that the heat pumps **120** only heat cold water. Further, the swing tank **150** provides extra capacity beyond the capabilities of the heat pumps **120** for extreme draws (i.e., extreme water usage in the building).

[0084] Containing the system **200** within the skids **210** and **220** allow for easy delivery and installation of a large-scale water heating system, i.e., the skids **210** and **220** allow the system **200** to be easily transported and integrated. The skids **210** and **220** are delivered pre-engineered and pre-packaged to the site of the building when the skids **210** and **220** are ready to be installed.

[0085] FIG. 3D illustrates a top view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. FIG. 3E illustrates a front view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. FIG. 3F illustrates a right side view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. In one embodiment, an exterior of a top side panel **210T** of the primary skid **210** includes one or more lift points **211** (e.g., rigging eyelets with substantially about 1 inch through holes). For example, in one embodiment, each corner of the exterior of the top side panel **210T** includes a lift point **211** positioned within proximity of the corner, as shown in FIG. 3D. The lift points **211** allow the entire primary skid **210** to be rigged such that a crane or a forklift can lift the primary skid **210** and drop-in the primary skid **210** into position. The primary skid **210** may be positioned onto a flat surface (such as on the rooftop of the building or on the ground adjacent to the building) and adjacent to the secondary skid **220**.

[0086] FIG. 3G illustrates a left side view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. In one embodiment, the first power connection panel **230** is mounted onto an exterior of the primary skid **210** (e.g., on an exterior of a side panel of the primary skid **210**, such as a left side panel **210B**). In one embodiment, the building water connections **250** include the hot water recirculation return **250A** for returning hot water to the system **200**, the cold water inlet **250B** for receiving cold water from the building, and the hot water outlet **250C** for supplying hot water to the building.

[0087] FIG. 3H illustrates a front view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. In one embodiment, the piping connections **290** include, but are not limited to, the following: a heat pump condensate drain **290B** (e.g., a ½ inch PVC stub) for

drainage of condensation from the first array of heat pumps **120** on the primary skid **210**, and tank temperature and pressure drains **290A** and **290C** (e.g., $\frac{3}{4}$ inch copper stubs) for drainage of any water leaks from each tank **140**, **150** (FIG. 3C) in the primary skid **210** after all excess air escapes from the tank **140**, **150** via its temperature and pressure relief port **147**, **153** (FIG. 3C).

[0088] FIG. 3I illustrates a right side view of the primary skid **210** of the system **200** in FIG. 3A, in one or more embodiments. In one embodiment, the first heat pump control terminal box **260** is mounted onto an exterior of the primary skid **210** (e.g., on an exterior of a side panel of the primary skid **210**, such as a right side panel **210C**). In one embodiment, the inter-skid piping connections **280** include a first heat pump manifold connection **280A** (e.g., a 1 inch Female National Pipe Tapered (FNPT) union) for cold water, a second heat pump manifold connection **280B** (e.g., a 1 inch FNPT union) for hot water, a cold water supply connection **280C** (e.g., from the cold water inlet **250B**) (e.g., a 2 inch FNPT union) to the secondary skid **220**, and a storage tank mid-array connection **280D** (e.g., for interconnecting the first storage tank **140** in the primary skid **210** with the third storage tank **140** in the secondary skid **220**) (e.g., a 2 inch FNPT union).

[0089] Any dimensions shown in FIGS. 3D-3E and 3G-3I are provided as examples only; the dimensions, size, and/or thickness of the skid **210**, water connections **250**, piping connections **290**, inter-skid piping connections **280**, and/or other components of the system **200** are not limited to the dimensions shown in these figures. In one embodiment, a height of the skid **210** is substantially about 103 $\frac{1}{4}$ inches, a length of the skid **210** ranges from substantially about 144 inches to substantially about 149 inches, and a width of the skid **210** ranges from substantially about 75 $\frac{3}{8}$ inches to substantially about 94 inches, as shown in FIGS. 3D-3E.

[0090] In one embodiment, vertical spacing between the water connections **250** and the flat surface/ground is substantially about 7 $\frac{1}{2}$ inches, horizontal spacing between the hot water recirculation return **250A** and the cold water inlet **250B** is substantially about 6 $\frac{1}{2}$ inches, horizontal spacing between the cold water inlet **250B** and the hot water outlet **250C** is substantially about 8 $\frac{1}{4}$ inches, and horizontal spacing between the hot water outlet **250C** and a corner base plate **212A** is substantially about 14 inches, as shown in FIG. 3G.

[0091] In one embodiment, vertical spacing between the piping connections **290** and the flat surface/ground is substantially about 9 inches, horizontal spacing between the tank temperature and pressure drain **290A** and the heat pump condensate drain **290B** is substantially about 6 inches, horizontal spacing between the heat pump condensate drain **290B** and the tank temperature and pressure drain **290C** is substantially about 5 $\frac{5}{8}$ inches, and horizontal spacing between the tank temperature and pressure drain **290A** and a corner base plate **212A** is substantially about 63 $\frac{1}{4}$ inches, as shown in FIG. 3H.

[0092] In one embodiment, vertical spacing between a topmost row of the inter-skid piping connections **280** (e.g., heat pump manifold connections **280A** and **280B**) and the flat surface/ground is substantially about 86 $\frac{1}{2}$ inches, vertical spacing between a bottommost row of the inter-skid piping connections **280** (e.g., cold water supply connection **280C**, storage tank mid-array connection **280D**) and the flat surface/ground is substantially about 78 $\frac{3}{8}$ inches, horizontal spacing between the heat pump manifold connections **280A** and **280B** is substantially about 5 $\frac{1}{2}$ inches, horizontal spacing between the cold water supply connection **280C** and the storage tank mid-array connection **280D** is substantially about 9 $\frac{1}{2}$ inches, horizontal spacing between the first heat pump manifold connection **280A** and a corner base plate **212A** is substantially about 17 $\frac{3}{8}$ inches, and horizontal spacing between the cold water supply connection **280C** and the same corner base plate **212A** is substantially about 17 $\frac{7}{8}$ inches, as shown in FIG. 3I.

[0093] FIG. 3J illustrates a top view of the secondary skid **220** of the system **200** in FIG. 3A, in one or more embodiments. FIG. 3K illustrates a front view of the secondary skid **220** of the system **200** in FIG. 3A, in one or more embodiments. FIG. 3L illustrates a left side view of the secondary skid **220** of the system **200** in FIG. 3A, in one or more embodiments. In one embodiment, an exterior of a top side panel **220T** of the secondary skid **220** includes one or more lift points **221** (e.g., rigging

eyelets with substantially about 1 inch through holes). For example, in one embodiment, each corner of the exterior of the top side panel **220T** includes a lift point **221** positioned within proximity of the corner, as shown in FIG. **3J**. The lift points **221** allow the entire secondary skid **220** to be rigged such that a crane or a forklift can lift the secondary skid **220** and drop-in the secondary skid **220** into position. The secondary skid **220** may be positioned onto a flat surface (such as on the rooftop of the building or on the ground adjacent to the building) and adjacent to the primary skid **210**.

[0094] FIG. **3M** illustrates a left side view of the secondary skid **220** of the system **200** in FIG. **3A**, in one or more embodiments. In one embodiment, the second heat pump control terminal box **270** is mounted onto an exterior of the secondary skid **220** (e.g., on an exterior of a side panel of the secondary skid **220**, such as a left side panel **220B**). In one embodiment, the inter-skid piping connections **285** include a first heat pump manifold connection **285A** (e.g., a 1 inch FNPT union) for cold water, a second heat pump manifold connection **285B** (e.g., a 1 inch FNPT union) for hot water, a cold water supply connection **285C** (e.g., from the cold water inlet **250B** in the primary skid **210**) (e.g., a 2 inch FNPT union), and a storage tank mid-array connection **285D** (e.g., for interconnecting the first storage tank **140** in the primary skid **210** with the third storage tank **140** in the secondary skid **220**) (e.g., a 2 inch FNPT union).

[0095] FIG. **3N** illustrates a front view of the secondary skid **220** of the system **200** in FIG. **3A**, in one or more embodiments. In one embodiment, the piping connections **295** include, but are not limited to, the following: a heat pump condensate drain **295B** (e.g., a ½ inch PVC stub) for drainage of condensation from the second array of heat pumps **120** on the secondary skid **220**, and tank temperature and pressure drains **295A** and **295C** (e.g., ¾ inch copper stubs) for drainage of any water leaks from each tank **140** (FIG. **3C**) in the secondary skid **220** after all excess air escapes from the tank **140** via its temperature and pressure relief port **147** (FIG. **3C**).

[0096] Any dimensions shown in FIGS. **3K-N** are provided as examples only; the dimensions, size, and/or thickness of the skid **220**, piping connections **295**, inter-skid piping connections **285**, and/or other components of the system **200** are not limited to the dimensions shown in these figures. In one embodiment, a height of the skid **220** is substantially about 103¼ inches, a length of the skid **220** ranges from substantially about 144 inches to substantially about 149 inches, and a width of the skid **220** ranges from substantially about 75⅜ inches to substantially about 94 inches, as shown in FIGS. **3K-3L**.

[0097] In one embodiment, vertical spacing between a topmost row of the inter-skid piping connections **285** (e.g., heat pump manifold connections **285A** and **285B**) and the flat surface/ground is substantially about 86½ inches, vertical spacing between a bottommost row of the inter-skid piping connections **285** (e.g., cold water supply connection **285C**, storage tank mid-array connection **285D**) and the flat surface/ground is substantially about 78⅜ inches, horizontal spacing between the heat pump manifold connections **285A** and **285B** is substantially about 6 inches, horizontal spacing between the cold water supply connection **285C** and the storage tank mid-array connection **285D** is substantially about 9½ inches, horizontal spacing between the first heat pump manifold connection **285A** and a corner base plate **212A** is substantially about 17⅜ inches, and horizontal spacing between the cold water supply connection **285C** and the same corner base plate **212A** is substantially about 17⅞ inches, as shown in FIG. **3M**.

[0098] In one embodiment, vertical spacing between the piping connections **295** and the flat surface/ground is substantially about 9 inches, horizontal spacing between the tank temperature and pressure drain **295A** and the heat pump condensate drain **295B** is substantially about 6 inches, horizontal spacing between the heat pump condensate drain **295B** and the tank temperature and pressure drain **295C** is substantially about 5⅞ inches, and horizontal spacing between the tank temperature and pressure drain **295A** and a corner base plate **212A** is substantially about 63¼ inches, as shown in FIG. **3N**.

[0099] FIG. **3O** illustrates an underside of each skid **210**, **220** of the system **200** in FIG. **3A**, in one

or more embodiments. In one embodiment, one or more tie down points **212** extend from an underside **210U** of each skid **210, 220** for securing the skid **210, 220** onto the surface. As shown in FIG. 3O, in one embodiment, the tie down points **212** include corner base plates **212A** at corners of the underside **210U**, side mounts **212B** at a first pair of opposing sides of the underside **210U**, and side mounts **212C** at a second pair of opposing sides of the underside **210U**. Each corner base plate **212A** and each side mount **212B, 212C** includes holes through which bolts/fasteners (e.g., 9/16 inch anchor bolts) are inserted to secure the skid **210, 220** to the flat surface/ground.

[0100] After the skids **210** and **220** are secured, the water connections **250** are connected to the building's water lines, the power connection panels **230** and **240** are connected to the building's main power supply, and the heat pump control terminal boxes **260** and **270** are powered on to begin running and starting the system **200**. The system **200** is a plug-and-play drop-in hot water solution.

[0101] Any dimensions shown in FIG. 3O are provided as examples only; the dimensions, size, and/or thickness of the skids **210** and **220** and/or other components of the system **200** are not limited to the dimensions shown in these figures. In one embodiment, each corner base plate **212A** is substantially about 10 inches by 10 inches, a length of each side mount **212B** is substantially about 46 inches, and a length of each side mount **212C** is substantially about 28 inches, as shown in FIG. 3O. In one embodiment, horizontal spacing between a corner base plate **212A** and a side mount **212B** along a width of a skid **210/220** is substantially about 4 11/16 inches, and horizontal spacing between a corner base plate **212A** and a side mount **212C** along a length of a skid **210/220** is substantially about 48 inches, as shown in FIG. 3O.

[0102] In one embodiment, each skid **210, 220** is a fully insulated enclosure (i.e., the steel frame of the skid **210** includes insulated side panels **210A, 210B, 210C, 210T**, etc.; the steel frame of the skid **220** includes insulated side panels **220A, 220B, 220C, 220T**, etc.), thereby allowing the system **200** to perform better than water heaters placed in mechanical rooms.

[0103] In one embodiment, the system **200** includes a master controller integrated in or coupled to the first heat pump control terminal box **260**. In one embodiment, the master controller is integrated with wireless communication options (e.g., WiFi, LTE, BMS).

[0104] Table 1 below provides example technical specifications for the system **200**.

TABLE-US-00001 TABLE 1 PERFORMANCE TOTAL HEAT PUMP CAPACITY [MBH] 123
TOTAL PRIMARY STORAGE VOLUME (GAL.) 660 MAX. DAILY HW OUTPUT
(GAL./DAY)* 2,560 MAX. DAILY RUNTIME (hr./DAY) 16 NOMINAL HOURLY HOT WATER
OUTPUT (GAL./hr.) 160 RECIRCULATION PUMP OUTPUT (HP) ½ RECIRC. PUMP MAX.
HEAD (FT.-H2O) 55 NOMINAL OPERATING SOUND PRESSURE LEVEL (dBA) 40 MAX.
SYSTEM WATER PRESSURE (PSI) 94 ELECTRICAL PRIMARY SECONDARY SKID PANEL
SKID PANEL SUPPLY 208 V/3 208 V/3 Ph/60 Hz Ph/60 Hz TOTAL 24.7 8.0 CONNECTED
LOAD [kVA] FULL LOAD 68.4 22.0 AMPS [A] MINIMUM 89.4 32.1 CIRCUIT AMPACITY
[A] MAXIMUM 90 35 OVERCURRENT PROTECTION [A] WEIGHT (APPROX.)
PRIMARY/SECONDARY DRY (LBS.) 5,500/4,900 PRIMARY/SECONDARY WETTED (LBS.)
8,400/8,700 COMPONENTS (8) HEAT PUMP REFRIGERANT TYPE: R-744 (CO2) WATER
HEATERS TOTAL CHARGE: 203 OZ. AIRFLOW (CFM): 4,800 EXTERNAL STATIC
PRESSURE (IN.-H2O): 0 (3) 220 GAL. STORAGE TANKS (1) SWING TANK, 119 GAL./12 kW
(2) EXPANSION TANKS, 45 GAL./30 GAL. ACCEPTANCE (1) ELECTRONIC 1 IN. BODY: 3-
94 GPM MIXING VALVE RECOMMENDED FLOW RANGE (1) RECIRCULATION PUMP,
ECM (1) HEAT PUMP STAGING CONTROLLER (1) INCOMING COLD WATER WYE-
STRAINER, 20-MESH CONSTRUCTION ENCLOSURE R8 INSULATION PANELS POWDER
COAT FINISH PIPING TYPE K COPPER DIELECTRIC CONNECTIONS WHERE REQUIRED
1-½" INSULATION, MIN. R-VALUE: 8.6 UL508A - LISTED POWER DISTRIBUTION
ELECTRICAL PANELS EQUIPMENT/STAGING CONTROLLER PAINTED CARBON STEEL
FRAME ELECTRICAL DISTRIBUTION PANEL WITH MAIN DISCONNECT INDIVIDUAL
COMPONENT DISCONNECTS ON HEAT PUMPS AND SWING TANK HEAT PUMPS

MOUNTED WITH VIBRATION ISOLATORS INSULATED BASE NOT PROVIDED BELOW
STORAGE AND SWING TANKS REVERSE-RETURN HEAT PUMP MANIFOLD PIPING
CONFIGURATION INDIVIDUAL HEAT PUMP SOLATION, DRAIN AND CHECK VALVE
*100° F. LIFT AND 16 hr./DAY MAX. COMPRESSOR RUNTIME

[0105] In one embodiment, each skid **210, 220** includes a service access door **212** (FIG. 3B), **222** (FIG. 3B) providing a user (e.g., a service technician) with access to components inside the skid **210, 220**.

[0106] Each HPWH system **100, 200** is a modular, pre-engineered, and pre-fabricated all-in-one centralized heat pump water heating system that is designed to integrate primary water heating and temperature maintenance, heat pumps, power landing, and controls all within a single, compact, one-piece or two-piece delivery unit. Each HPWH system **100, 200** is ready to operate after an installer plugs-and-plays the system **100** or **200** with an on-site's (e.g., a multi-family residential building) power and water connections. Utilizing heat pumps and stratified storage tanks, this plug-and-play approach greatly simplifies installation and addresses space constraints in various settings (e.g., urban settings where space is at a premium). The components of each system **100, 200** can be installed and started-up in as little as a day, thereby shortening the length of time it takes to custom design and build equivalent systems from months to a single day.

[0107] In one embodiment, each modular HPWH system **100, 200** is a pre-manufactured drop-in, central heat pump plant that uses CO₂ as a refrigerant. Each modular system **100, 200** is ozone-friendly and does not contribute to global warming. Each modular system **100, 200** is designed and fabricated off-site, then delivered when needed. After each skid of the modular system **100** or **200** is physically secured on-site, the only connections required are mains power and the following water connections: cold water in, hot supply water, and recirculation.

[0108] In one embodiment, modular HPWH systems of different sizes and/or configurations than the modular systems **100** and **200** shown in FIGS. 1A-1C and 3A-3O are provided. Each modular HPWH system of a particular size is a complete central heat pump water heater and includes all heat pumps, tanks, pumps, controls, power panel, mixing station, piping, insulation, all housed in a seismic rated steel frame with powder coat finish that is enclosed in an insulated enclosure made of powder coated aluminum panels with integrated rigid R-8 insulation (i.e., a skid). Each skid of a modular system can be lifted by a forklift or craned in place. Sizes of each skid of a modular system are determined by a combination of heat pump capacity and water storage. These combinations match up to typical residential building loads. Engineers can review building load sizes by common industry sizing tools and select a skid to match the building load. Multiple skids can be used in zones on a single building to reach larger loads.

[0109] In one embodiment, each skid of a modular system is to be mounted on an elevated concrete pad or other suitable location. Once located, the skid is to be anchored according to a provided Installation, Operation, and Maintenance (IOM) manual.

[0110] Skids contain a power connection panel for the electrical supply to the building as well as building plumbing/water connections consisting of hot water supply to building, cold water make-up from building, and recirculation return from building.

[0111] At least one skid of a modular system has a power connection panel which requires external connection. For example, a skid may require 208V/3 Ph/60 Hz consisting of five wires: Lines A, B, C, Ground, and Neutral. Wire gauge and other details are to be determined by an installer based on NEC and local code requirements. Upstream overcurrent protection (e.g., building circuit breaker) is to be verified not to exceed the Maximum Overcurrent Protection (MOCP) listed on the nameplate of the distribution panel. As another example, each of the skid **210, 220** has a heat pump control terminal box **260, 270** with all required control wiring terminated-this box will need a bridge wiring connection to allow inter-skid communication and controls. Wiring and conduit requirements will conform with NEC and local codes.

[0112] In one embodiment, each modular HPWH system **100, 200** includes a staging controller that

has primary enable/disable capability via its local user interface with secondary capability via Building Management System (BMS) command. When enabled, the controller will operate under its own internal controls to stage heat pumps and monitor for alarm conditions. The controller's Enable/Disable is stored in the retained memory to allow the modular system **100** or **200** to automatically restart after a building power failure. To prevent the controller from calling heat pumps to start after a power cycle, place it in Off mode before restarting. The controller shall monitor the Tank Temperature and HPWH Inlet water temperature to determine call for heating. The controller shall employ minimum On, Off, and Consecutive Start safety timers to prevent undesired heat pump cycling. The controller shall balance unit run times by starting the units in order of lowest operating hours. In the event that a unit is replaced, the run hour counters can be individually reset. The controller will monitor outdoor air temperature via its Outside Air temperature sensor and activate freeze protection when the temperature drops below -20° F. In the event a heat pump experiences a critical fault, it will report this to the controller via its alarm contact closure.

[0113] Each skid of a modular system is pneumatically pressure-tested prior to leaving the factory, but some leaks may still be present. Fittings have been located throughout each skid to aid in air purging. Each outlet (upper connection) of a heat pump has a ball/drain combination valve to allow a garden hose connection to flush the internals of the heat pump. Mixing valve outlet also contains a ball/drain combination valve with garden hose outlet. Recirculation pump outlet flange has an integrated isolation/drain valve with garden hose outlet for pump priming. Each storage tank and swing tank has an automated air release valve.

[0114] In another embodiment, each modular HPWH system **100**, **200** uses another low GWP (Global Warming Potential) refrigerant other than CO₂ that could offer similar benefits but with different operational characteristics or cost structures.

[0115] In another embodiment, modular systems of different modular configurations that offer more customizable options in terms of size, capacity, and component layout are provided to cater to a broader range of building types or specific industry needs.

[0116] In another embodiment, a design can be made with PCM (Phase Change Material) or alternative thermal storage solutions. This might include advanced sensible heat storage materials or other types of phase change materials with different melting points and thermal properties.

[0117] In another embodiment, a modular system designed to integrate more seamlessly with renewable energy sources like solar or wind power is provided, optimizing energy use for heating water during peak production times.

[0118] In another embodiment, a modular system provided is enhanced with more advanced IoT (Internet of Things) capabilities, AI-based predictive maintenance, and remote monitoring. This would offer improved user experience, energy management, and maintenance scheduling.

[0119] In another embodiment, a modular system provided is maintenance-friendly, with easily replaceable components or designs that allow for future technological upgrades without replacing the entire system.

[0120] In another embodiment, a modular system provided is manufactured using alternative manufacturing techniques or materials that reduce the overall cost of the system without significantly impacting performance.

[0121] In another embodiment, a modular system provided has a more user-friendly interface (e.g., with touchscreen controls or more intuitive settings) to appeal to users who prioritize ease of use.

[0122] References in the claims to an element in the singular is not intended to mean “one and only” unless explicitly so stated, but rather “one or more.” All structural and functional equivalents to the elements of the above-described exemplary embodiment that are currently known or later come to be known to those of ordinary skill in the art are intended to be encompassed by the present claims. No claim element herein is to be construed under the provisions of 35 U.S.C. section 112, sixth paragraph, unless the element is expressly recited using the phrase “means for”

or “step for.”

[0123] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0124] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the embodiments has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention.

[0125] Though the embodiments have been described with reference to certain versions thereof; however, other versions are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the preferred versions contained herein.

Claims

1. A compact and centralized heat pump water heater system comprising: a modular skid including: a stratified primary storage tank for storing cold water and hot water, wherein the storage tank is disposed inside the skid; and an array of heat pumps for heating the cold water into the hot water, wherein the heat pumps are mounted onto an exterior of the skid; and a master controller for controlling the system; wherein the hot water is supplied to a building structure.
2. The system of claim 1, wherein the building structure is one of a multi-family residential building, an office, a factory, an industrial space, or a commercial space.
3. The system of claim 1, wherein the modular skid further includes: a recirculation loop; and a swing tank, a master mixing valve, and a recirculation pump for maintaining a target temperature of the hot water in the recirculation loop.
4. The system of claim 1, wherein the modular skid further includes: a plurality of water connections for connecting to a water supply of the building structure; and a power connection panel for connecting to a power supply of the building structure.
5. The system of claim 4, wherein the system is delivered pre-engineered and pre-manufactured to a site of the building structure for installation.
6. The system of claim 5, wherein the installation comprises: connecting the water connections to the water supply of the building structure; connecting the power connection panel to the power supply of the building structure; and powering on the master controller to start and run the system.
7. The system of claim 1, wherein the modular skid further includes: a plurality of piping connections for drainage.
8. The system of claim 1, wherein each heat pump uses carbon dioxide as a refrigerant.
9. The system of claim 1, where the array comprises an even number of heat pumps.
10. A compact and centralized heat pump water heater system comprising: a modular primary skid including: a first stratified storage tank for storing cold water and hot water, wherein the first stratified storage tank is disposed inside the primary skid; and a first array of heat pumps for heating the cold water into the hot water, wherein the first array of heat pumps are mounted onto an exterior of the primary skid; a modular secondary skid including: at least one additional stratified storage tank for storing the cold water and the hot water, wherein the at least one additional stratified storage tank is disposed inside the secondary skid; and a second array of heat pumps for

heating the cold water into the hot water, wherein the second array of heat pumps are mounted onto an exterior of the secondary skid; and a master controller for controlling the system; wherein the hot water is supplied to a building structure.

11. The system of claim 10, wherein the building structure is one of a multi-family residential building, an office, a factory, an industrial space, or a commercial space.

12. The system of claim 10, wherein the primary skid further includes: a recirculation loop; and a swing tank, a master mixing valve, and a recirculation pump for maintaining a target temperature of the hot water in the recirculation loop.

13. The system of claim 10, wherein: the primary skid further includes: a plurality of water connections for connecting to a water supply of the building structure; and a first power connection panel for connecting to a power supply of the building structure; and the secondary skid further includes: a second power connection panel for connecting to the power supply of the building structure.

14. The system of claim 13, wherein the system is delivered pre-engineered and pre-manufactured to a site of the building structure for installation.

15. The system of claim 14, wherein the installation comprises: connecting the water connections to the water supply of the building structure; connecting the first power connection panel and the second power connection panel to the power supply of the building structure; and powering on the master controller to start and run the system.

16. The system of claim 10, wherein each primary and secondary skid further includes: a plurality of piping connections for drainage.

17. The system of claim 10, wherein each heat pump uses carbon dioxide as a refrigerant.

18. The system of claim 10, where each array comprises an even number of heat pumps.

19. The system of claim 10, wherein the first array comprises a different number of heat pumps than the second array.

20. The system of claim 10, wherein each primary and secondary skid further includes: a plurality of inter-skid piping connections for interconnecting the primary skid with the secondary skid.
